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The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2022

COURSE CODE: AS 220

COURSE NAME: PRINCIPLES OF CROP SCIENCE

TIME ALLOWED: THREE (3) HOURS

DATE: 17TH OCTOBER 2022

INSTRUCTIONS: 1. Answer the ALL the questions

2. The exam has a total of 60 marks

Course Code: AS 220 (Principles of Crop Science)

Q1. A farmer comes to you for consultation on the crop to grow this farming season of 2022/2023. After considering the weather pattern, you advise the farmer to grow a **fiber** crop. What **two** crops can you recommend and what **three** products can be produced from fiber crops. **(5 marks)**

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Q2.Describe **light** and **carbon** (dark) reactions that take place in the plant. **(5 marks)**

Q3. During the rainy season it was discovered that **sunhemp** seed that accidentally dropped in the garden has germinated. The owner of the garden wants to remove the germinated plants immediately, whatadvice can you give to the owner of the garden? **(2 marks)**

Q4. Eco-petroleum company is planning to produce maize for the purpose of producing biofuel. Why should the production be promoted and why should the production be discouraged? **(3 marks)**

Q5.Mr. Mwachindalo keeps livestock in an open area and one day when the sun was still shining, he decided to put fruit tree seedlings in the same area with livestock.As a crop science student how would you support the benefit of plants and animals in terms of food production and utilization? **(4 marks)**

Q6. A farmer in Lunte district measured that each maize plant in the field occupied the space of **2,250cm^2**. Using this information answer the following questions.

a) Calculate the spacing between the plants (intra-row spacing) if the spacing between beds is at 90cm. **(4 marks)**

b) The total maize field was measured to be **6,750m^2**, how many plants are expected to be in the field? **(3 marks)**

c) The farmer bought **90 X 50kg** bags of fertilizer to apply in that same field. In order to achieve even distribution of the fertilizer, how much fertilizer can be recommended to apply per plant? **(4 marks)**

Q7. Apart from human consumption, what else can the maize crop be used for? **(3 marks)**

Q8. What is relay cropping? **(3 marks)**

Q9. Both manure and inorganic fertilizers are used in crop production, how do they differ? **(4 marks)**

Q10. Why is it beneficial to have a leaf of the crop that is thick than the thin leaf? **(2 marks)**

Q11. Name parts of the crop plant that can be a source of producing assimilates? **(3 marks)**

Q12. After planting beans for four weeks, you confirm that the bacteria rhizobia have nodulated the crop. What simple test can you carry out to check if the rhizobium is efficient or not? **(4 marks)**

Q13. Cabbage crop was infested with aphids and one of the workers at the farm over applied thechemical abamectin. Later it was discovered that enzymes were denatured. What processes in the plant do you expect to be affected by the scorching of the crop? **(3 marks)**

Q14. Explain the process of respiration in the plant. **(4 marks)**

Q15 Differentiate micro nutrients from macro nutrients? **(4 marks)**

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